4.2.1 Fertility and assisted reproduction

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the parts of the male and female urinogenital systems	To include the interpretation of diagrams, photomicrographs and electron micrographs. *M0.1, M0.2, M1.1, M1.2, M1.8*
(b)	**(i)** the histology of the ovaries and testes	To include the interpretation of diagrams, photomicrographs and electron micrographs. *M0.1, M0.2, M1.1, M1.2, M1.8*
	(ii) observations of the histology of the ovaries and testes made using the light microscope	**PAG1** HSW4, HSW5, HSW6
(c)	the process of gametogenesis	To include oogenesis and spermatogenesis and relate these to mitosis and meiosis.
(d)	the structure of the secondary oocyte and sperm related to their functions	To include the interpretation of diagrams, photomicrographs and electron micrographs. *M0.1, M0.2, M1.1, M1.2, M1.8*
(e)	the role of hormones in gametogenesis	To include spermatogenesis: FSH, GnRH, LH, testosterone and inhibin **AND** oogenesis: FSH, LH, progesterone, oestrogen and GnRH.
(f)	the role of hormones in the regulation of the menstrual cycle	To include LH, FSH, progesterone and oestrogen.

(g)	the process of fertilisation	To include the importance in restoring chromosome number and introducing genetic variation.
(h)	the use of monoclonal antibodies in pregnancy testing	No details of the production of monoclonal antibodies are required.
(i)	a consideration and evaluation of the biological and ethical issues surrounding infertility	To include the production of abnormal sperm, low sperm count and blocked vas deferens, ovulatory disorders, oviduct blockage, endometriosis and anti-sperm antibodies. *M0.1, M0.2, M1.1, M1.2, M1.8*
(j)	assisted reproduction	Treatments and techniques to include intrauterine insemination, IVF, intra-cytoplasmic sperm injection (ICSI), donor sperm insemination, surgical sperm retrieval, ovulation induction, frozen embryo replacements and gamete intra-fallopian transfer. HSW9, HSW10, HSW12